

01__Basic Calculus | Notations

What Is Calculus?

Calculus is a branch of mathematics concerned with the study of **change** and **accumulation**. It provides a formal framework for analyzing how quantities vary with respect to one another and for computing quantities such as rates, slopes, areas, and total effects.

Calculus is traditionally divided into two main branches:

1. **Differential Calculus**
2. **Integral Calculus**

These branches are linked by the **Fundamental Theorem of Calculus**, which establishes the relationship between differentiation and integration.

1. Differential Calculus

Differential calculus focuses on the concept of the **derivative**, which measures how a function changes as its input changes.

Derivative

Given a function $f(x)$, its derivative at a point x is defined as

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

The derivative represents:

- The **instantaneous rate of change** of the function
- The **slope of the tangent line** to the graph at a given point

Applications include:

- Motion (velocity and acceleration)
- Optimization problems
- Modeling change in physical, biological, and economic systems

2. Integral Calculus

Integral calculus deals with the concept of **accumulation** and the computation of quantities such as area and total change.

Integral

The definite integral of a function $f(x)$ over an interval $[a, b]$ is defined as

$$\int_a^b f(x) dx$$

It represents:

- The **area under the curve** between a and b
- The **total accumulation** of the quantity described by $f(x)$

Indefinite integrals correspond to families of antiderivatives:

$$\int f(x) dx = F(x) + C$$

where $F'(x) = f(x)$.

3. Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus connects differentiation and integration.

Part I:

If $F(x) = \int_a^x f(t) dt$, then

$$F'(x) = f(x)$$

Part II:

If $F'(x) = f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a)$$

This theorem shows that integration and differentiation are inverse processes.

4. Limits and Continuity

The concept of a **limit** underlies all of calculus.

$$\lim_{x \rightarrow a} f(x)$$

Limits are used to:

- Define derivatives
- Define integrals
- Analyze continuity and behavior near points

A function is **continuous** at a point if the limit exists and equals the function value at that point.

5. Multivariable Calculus (Brief Overview)

When functions depend on more than one variable, calculus extends to higher dimensions.

Key concepts include:

- Partial derivatives
- Gradients
- Multiple integrals

These tools are essential in physics, engineering, economics, and modern applied fields.

Why calculus matters (especially today)

Calculus is the math behind:

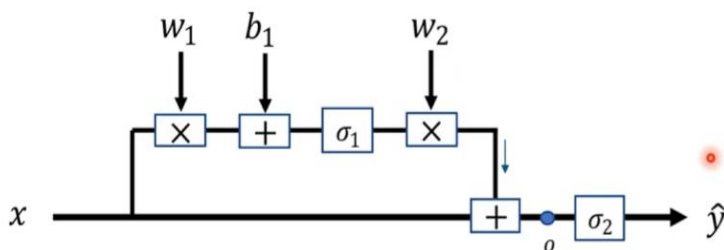
- Physics and engineering
- Economics and finance
- **Machine Learning & AI**
- Data science and optimization

In ML:

- Derivatives → **learning**
- Gradients → **direction to improve**
- Optimization → **better models**

No calculus = models can't learn.

Importance:



Let $\sigma'(x)$ be the derivative of $\sigma(x)$, which are defined as

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\text{Loss} = \|\hat{y} - y\|_2^2 = (\hat{y} - y)^2$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$

Solution:

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y)$$

$$\frac{\partial L}{\partial o} = \frac{\partial L}{\partial \hat{y}} \sigma'_2(x + w_2 \sigma_1(w_1 x + b_1))$$

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial o} \sigma_1(w_1 x + b_1)$$

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial o} w_2 \sigma'_1(w_1 x + b_1)$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial o} w_2 \sigma'_1(w_1 x + b_1) x$$

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial o} + \frac{\partial L}{\partial o} w'_2(w_1 x + b_1) w_1$$

$$\Rightarrow \frac{\partial L}{\partial \hat{y}_t} = 2(\hat{y}_t - y_t)$$

$$\Rightarrow \frac{\partial L}{\partial o_t} = \frac{\partial L}{\partial \hat{y}_t} \cdot \frac{\partial \hat{y}_t}{\partial o_t} + \frac{\partial L}{\partial h_t} \cdot \frac{\partial h_t}{\partial o_t} = \frac{\partial L}{\partial \hat{y}_t} + \frac{\partial L}{\partial h_t}$$

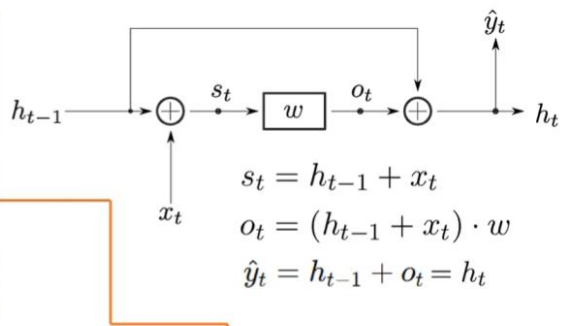
$$\Rightarrow \frac{\partial L}{\partial w_t} = \frac{\partial L}{\partial o_t} \cdot \frac{\partial o_t}{\partial w_t} = \frac{\partial L}{\partial o_t} \cdot (h_{t-1} + x_t) = s_t \cdot \frac{\partial L}{\partial o_t}$$

$$\Rightarrow \frac{\partial L}{\partial w} = \sum_t \frac{\partial L}{\partial w_t}$$

$$\Rightarrow \frac{\partial L}{\partial s_t} = \frac{\partial L}{\partial o_t} \cdot \frac{\partial o_t}{\partial s_t} = \frac{\partial L}{\partial o_t} \cdot w$$

$$\Rightarrow \frac{\partial L}{\partial x_t} = \frac{\partial L}{\partial s_t} \cdot \frac{\partial s_t}{\partial x_t} = \frac{\partial L}{\partial s_t}$$

$$\Rightarrow \frac{\partial L}{\partial h_{t-1}} = \frac{\partial L}{\partial o_t} + \frac{\partial L}{\partial s_t}$$



Derivative of function:

If you have a function $f(x)$, the **derivative** is written as:

- $f'(x) \leftarrow$ (Lagrange notation)
- $\frac{df}{dx} \leftarrow$ (Leibniz notation)
- $Df(x) \leftarrow$ (less common, operator notation)

They all mean: how much does $f(x)$ change as x changes?

Let's say:

$$f(x) = x^2$$

Then the derivative is:

- $f'(x) = 2x$
- Or: $\frac{d}{dx}(x^2) = 2x$

Partial Derivatives:

If $f(x, y)$, then:

- $\frac{\partial f}{\partial x}$ = derivative with respect to x
- $\frac{\partial f}{\partial y}$ = derivative with respect to y

Used a lot in ML when you have functions with many variables.

$$f(x, y) = 3x^2y + 2xy^2$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(3x^2y + 2xy^2)$$

$$\frac{\partial f}{\partial x} = 6xy + 2y^2$$

Partial Derivatives Orders:

- **First-Order** Partial Derivatives • $\frac{\partial f}{\partial x}$ • $\frac{\partial f}{\partial y}$

They show how the function changes as each variable changes on its own.

- **Second-Order** Partial Derivatives: These measure how the rate of change itself changes.

- Pure Second-Order Derivatives • $\frac{\partial^2 f}{\partial x^2}$ • $\frac{\partial^2 f}{\partial y^2}$

- Mixed Second-Order Derivatives • $\frac{\partial^2 f}{\partial x \partial y}$ • $\frac{\partial^2 f}{\partial y \partial x}$

For ML, Deep Learning, Optimization: $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$

Integral Calculus $\int$:

♦ Indefinite Integral (No Limits)

Represents the *antiderivative* (reverse of differentiation):

- $\int f(x) dx$

This gives a general function $F(x)$ such that $F'(x) = f(x)$

♦ Definite Integral (With Limits)

Represents the *area under the curve* from a to b :

- $\int_a^b f(x) dx$

This gives a number, not a function.

If $f(x) = 2x$:

- Indefinite integral:

$$\int 2x dx = x^2 + C \text{ (C = constant)}$$

- Definite integral:

$$\int_0^2 2x dx = [x^2]_0^2 = 4 - 0 = 4$$